

Qatar-2b

R-0688

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_260120 · created 2026-07-27T20:20:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461060.9826 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.027
Transit depth (Rp/Rs)^2	1.78 +/- 0.72 [%]
Orbital Inclination (inc)	89.63 +/- 1.64
Airmass coefficient 1 (a1)	0.941 +/- 0.025
Airmass coefficient 2 (a2)	0.041 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.63 %
Best Comparison Star	#1 - [45, 46]
Optimal Method	PSF photometry
Transit Duration (day)	0.0741 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.133 ± 0.027 vs 0.16327 (deviation 1.12σ)
Duration fit vs published	0.0741 d vs 0.0767 d (deviation 1.12σ)
Mid-time O-C	-0.5 min
Depth SNR	4.9
Residual scatter	2.63 %

Light curve

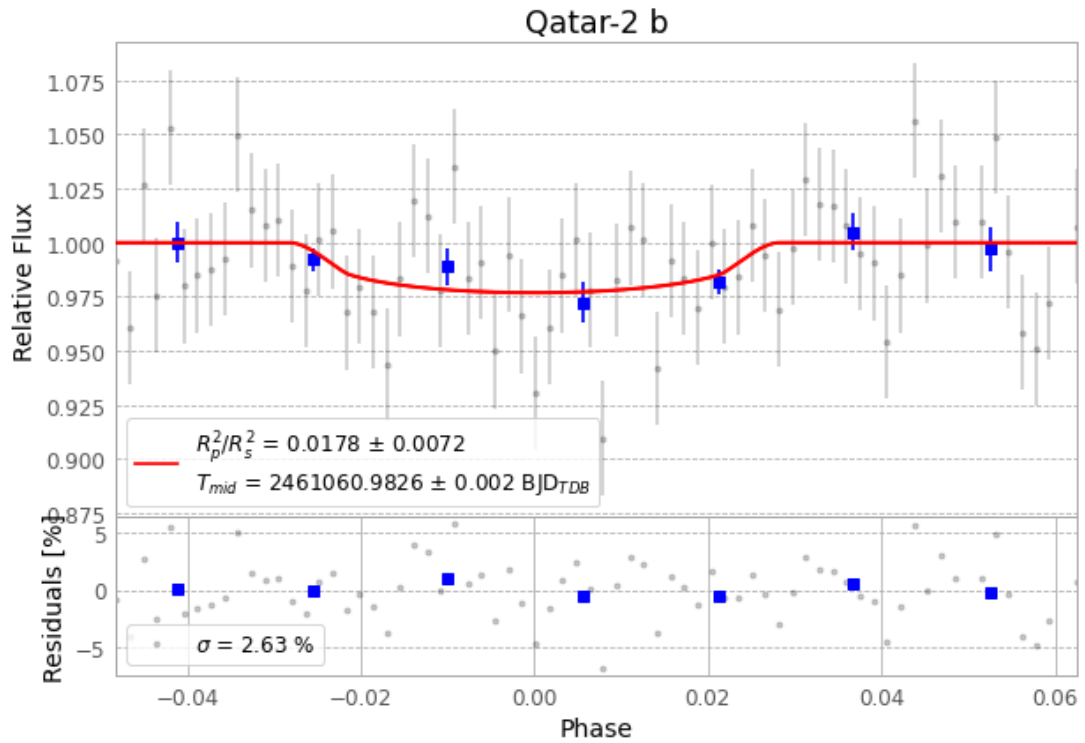


Figure 1 — FinalLightCurve_Qatar-2 b_2026-01-20.png (SHA-256 4839bc96035b5ed3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [505, 208], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 998c62700e625c1d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

71 FITS frames (47 MB), Qatar-2260120100015.FITS → Qatar-2260120133313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-260120.log (6f0d447ad1b0...), FinalLightCurve_Qatar-2 b_2026-01-20.png (4839bc96035b...), FinalLightCurve_Qatar-2 b_2026-01-20.csv (969de6f4f738...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:20Z.